Incident Response Investigation Report

Project: Splunk Incident Response Investigation Dashboard **Type:** Simulated cybersecurity lab

Event date: October 15, 2026 **Data source:** incident_response_logs.csv **Splunk index:** incident_response_lab

1. Executive Summary

A review of simulated Windows authentication events identified eight consecutive failed login attempts against account **j.smith** from source IP **203.0.113.25**, followed by a successful VPN login and subsequent application-server access. This pattern is consistent with password guessing or compromised credentials, but does not establish malicious intent or confirmed compromise. The investigation is a training exercise using example IP addresses, not a real-world security incident.

2. Scope and Method

Imported a CSV log dataset into Splunk, indexed events in **incident_response_lab**, and used SPL searches to aggregate Windows authentication event codes 4624 (successful logon) and 4625 (failed logon), correlate source IPs and accounts, and visualize activity across five dashboard panels. Event code 4634 (logoff) was used in the timeline view.

3. Evidence and Timeline

Time (Oct 15, 2026)	Observed event
09:05-09:12	Eight failed logins (4625) for j.smith from 203.0.113.25 on VPN-GW-01
09:13	Successful login (4624) for j.smith from 203.0.113.25 on VPN-GW-01
09:14	Successful authentication / application-server access (4624) on APP-SRV-01
09:16-09:17	Two failed logins (4625) for admin from 198.51.100.44
09:20	Logoff (4634) in the j.smith session timeline

4. Key Findings

Authentication totals: 15 events included 10 failed logins and 5 successful logins (66.7% failed). **Highest-volume source:** 203.0.113.25 generated eight failed attempts. **Additional activity:** 198.51.100.44 generated two failed attempts against admin. **Concerning sequence:** repeated failures, successful authentication, and application access in a short interval.

5. Assessment and Limitations

Assessment: Suspicious authentication sequence; further verification required. The dataset does not provide MFA results, endpoint telemetry, geolocation, authorized-user confirmation, or proof of data access/exfiltration. A successful login following failures is not, by itself, proof of a brute-force attack. The IPs shown are documentation/test ranges, reinforcing the simulated nature of this lab.

6. Recommended SOC Response

Validate whether the login was authorized with the account owner; review VPN, identity-provider and MFA logs; correlate endpoint and server activity around 09:13-09:20; inspect other accounts targeted by the source IPs; preserve relevant log evidence; and, if compromise is substantiated, follow the organization's incident-response playbook for session revocation, credential reset, containment, and escalation.

7. Dashboard Deliverables

Five panels: (1) Failed Login Attempts by IP, (2) Suspicious Login Activity Timeline, (3) Successful Logins After Failed Attempts, (4) Authentication Events by Status, and (5) Suspicious IP Investigation Details. Attach the two captured dashboard screenshots in the GitHub project README as visual evidence.

8. Skills Demonstrated

Splunk Enterprise; Search Processing Language (SPL); CSV ingestion and indexing; Windows event analysis; authentication-log correlation; suspicious-login triage; incident documentation; security dashboard design.

Portfolio note: This report is based on a controlled practice dataset. Findings are analytical observations, not a claim of responding to an actual production breach.